

STA 130A – Homework 2

Submission due: Thu, October 8 at 11:59 PM PT

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Instructions: Upload a single PDF file to Gradescope via Canvas (“Homework 2” under “Assignments”). Name the file using the prefix of your UC Davis email ID and the homework number (e.g., `dgsong_hw2.pdf`). Include “STA 130A,” your name, and the last four digits of your student ID on the front page. No late submissions will be accepted; any submission received after the deadline will receive 0 points. For full submission requirements and the late submission policy, see the syllabus.

Show your reasoning, not only final answers. For each PMF, specify its values on all possible values of the random variable and state that it is zero elsewhere. You may leave exact answers in terms of factorials, binomial coefficients, or finite sums. Problems 1–4 are required and total **100 points**. Problem 5 is optional and offers up to **10 additional bonus points**.

Problem 1. Independence (30 points in total).

- (a) **(10 points)** Let A and B be independent events. Prove that if $A \subseteq B$, then either $P(A) = 0$ or $P(B) = 1$.
- (b) **(10 points)** Two fair six-sided dice are rolled independently. Define the events

$$A = \{\text{the first die shows 6}\},$$

$$B = \{\text{the second die shows 6}\},$$

$$C = \{\text{the dice show the same number}\}.$$

Are A, B, C pairwise independent? Are they mutually independent? Justify both answers by calculating the relevant probabilities.

- (c) **(10 points)** Adapted from [BT08, Chapter 1, Problem 39, p. 62]. A professor gives a lecture if and only if at least k of the n enrolled students attend, where $1 \leq k \leq n$. The weather is bad with probability q , and good otherwise. Conditional on the weather, students attend independently: each attends with probability p_g in good weather and p_b in bad weather. Assume $0 < q, p_g, p_b < 1$. Find the probability that the professor gives the lecture, in terms of n, k, q, p_g, p_b .

Problem 2. Counting (20 points in total).

There are eight distinct volunteers: five experienced volunteers and three new volunteers. Explain the choices represented by each factor in your answers.

- (a) **(5 points)** Three ordered time slots each need one volunteer. How many schedules are possible when a volunteer may fill more than one slot? How many are possible when each volunteer may fill at most one slot?
- (b) **(5 points)** How many three-person teams can be selected from the eight volunteers? Team members have no distinct roles. Explain how your answer relates to the number of schedules with no repeated volunteer in part (a).
- (c) **(5 points)** A three-person team is selected uniformly from all three-person teams. Find the probability that it contains exactly two experienced volunteers. Obtain your answer in two ways: by counting teams, and by selecting three volunteers sequentially without replacement, choosing uniformly from the remaining volunteers at each step.
- (d) **(5 points)** All eight volunteers stand in a row. How many arrangements have no two new volunteers next to each other? *Hint:* First arrange the experienced volunteers. Where can the new volunteers be placed?

Problem 3. Discrete distributions (20 points in total).

- (a) **(10 points)** Adapted from [BT08, Chapter 2, Problem 7(a), p. 120]. You have five keys, exactly one of which opens a door. Let T be the number of trials needed to open the door, including the successful trial. Find the PMF of T under each of the following two strategies:
 - (i) After an unsuccessful trial, discard that key. At each trial, choose uniformly from the keys that remain.
 - (ii) At each trial, independently choose uniformly from all five keys, including any previously tried keys.
- (b) **(10 points)** The number N of messages received during an hour has a Poisson distribution with parameter 2. State its PMF and find $P(N \geq 2)$.

Problem 4. Functions, expectation, and variance (30 points in total).

- (a) **(10 points)** A student starts with 3 points and answers 3 true/false questions. Each answer is independently correct with probability $2/3$. Each correct answer earns 2 additional points, whereas each incorrect answer deducts 1 point. Let X denote the student's total number of points after all 3 questions have been answered. Find the PMF of X , its mean, and its variance.
- (b) **(10 points)** Adapted from [BT08, Chapter 2, Problem 16(a)–(c), p. 122]. A discrete random variable X has PMF

$$p_X(x) = \begin{cases} x^2/c, & x \in \{-3, -2, -1, 0, 1, 2, 3\}, \\ 0, & \text{otherwise,} \end{cases}$$

where $c > 0$ is a normalizing constant. Find c and $\mathbb{E}[X]$. Then find the PMF of $Z = (X - \mathbb{E}[X])^2$ and use it to calculate $\text{Var}(X)$.

- (c) **(10 points)** Let X be a discrete random variable with $\mathbb{E}[X^2] < \infty$, and let $a \in \mathbb{R}$. Prove that

$$\mathbb{E}[(X - a)^2] = \text{Var}(X) + (\mathbb{E}[X] - a)^2.$$

Determine the value of a that minimizes $\mathbb{E}[(X - a)^2]$.

Problem 5*. Optional bonus (up to 10 additional points).

Each part is optional and worth up to 5 bonus points.

- (a) **(5 bonus points)** Adapted from [BT08, Chapter 1, Problem 60, p. 69]. A standard deck of 52 distinct cards, including four aces, is shuffled uniformly at random and dealt to four players, with 13 cards per player. Find the probability that each player receives exactly one ace. Clearly identify the equally likely outcomes used in your counting argument.
- (b) **(5 bonus points)** Adapted from [BT08, Chapter 1, Problem 40, p. 62]. A coin comes up heads with probability p , where $0 < p < 1$, and successive tosses are independent. For each integer $n \geq 0$, let Q_n be the probability of an even number of heads in the first n tosses, so $Q_0 = 1$. Derive a recursion relating Q_n to Q_{n-1} , and solve it to show that

$$Q_n = \frac{1 + (1 - 2p)^n}{2}, \quad n \geq 1.$$

Find $\lim_{n \rightarrow \infty} Q_n$ and briefly interpret the result for a biased coin.

References

- [BT08] Dimitri P. Bertsekas and John N. Tsitsiklis. *Introduction to Probability*, 2nd edition. Athena Scientific, 2008.